Introduction to probability through problems

01.1 · Worked student edition

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A ticket-routing review

A service desk checks whether an LLM routes tickets to the appropriate teams. A reviewer supplies reference decisions; the model supplies classifications. What would it mean for them to agree?

We will compare the reviewer reference outcome with the model classification ticket by ticket. The example is hypothetical. The reviewer decisions are operational references, not claims of error-free ground truth.

Six reviewed tickets

For this hypothetical audit, the model uses a 0.50 threshold separately on each destination.

Ticket	Request	Reviewer	Model
t_1	Duplicate charge; cannot log in	Billing + Account Support	Billing
t_2	Duplicate monthly charge	Billing	Billing
t_3	MFA phone change; cannot log in	Account Support	Account Support
t_4	Delivery-status question	Other	Account Support
t_5	Duplicate renewal charge	Billing	Billing
t_6	Locked after password reset	Account Support	Account Support

What does the table count as an agreement?

For t_1, the model scores are 0.55 for Billing and 0.35 for Account Support. The reviewer reference includes both destinations. Exact agreement means the model classification equals the full reviewer reference outcome.

Which rows agree?

Mark exact agreement between the reviewer reference and model classification. How many of the six records agree?

Exact agreements occur on t_2, t_3, t_5, t_6. Thus the observed agreement fraction is $4/6 = 2/3$.

What does four out of six describe?

The table records four agreements. Is $4/6$ a probability for every question about this model?

It is first an observed proportion in these six records. It becomes the probability of agreement if the experiment is to select one of these six ticket IDs uniformly. A claim about future tickets needs a target population and assumptions linking this audit to it.

Equal-likelihood counting

If one of the six ticket IDs is selected uniformly, what is the probability that the selected record shows agreement?

The classical rule for a finite set of equally likely cases is $P(E) = |E|/|\Omega|$. Here each ID has probability $1/6$ and four IDs show agreement, so $P(\text{agreement}) = 4/6 = 2/3$. The uniform selection rule justifies equal weights for IDs in this experiment.

The cases being counted matter

The occupied reviewer/model pairs are: both destinations / Billing (t_1); Billing only / Billing (t_2, t_5); Account only / Account (t_3, t_6); Other / Account (t_4). If one pair type were selected uniformly, what agreement probability would result?

Two of four occupied types are agreements, so the answer under uniform selection among types is $2/4 = 1/2$. It differs from $4/6$ under uniform selection among ticket IDs because the categories contain unequal numbers of records.

A larger set of possible label pairs

There are four possible reviewer types and four model outputs (neither, Billing only, Account Support only, both). How many pairs are possible in their full Cartesian product? How many appear here?

There are $4 \times 4 = 16$ possible pairs in the full product. Four are occupied in this six-row table. Uniform weighting over 16 pairs would give $4/16 = 1/4$ agreement. This third selection rule is not implied by the audit.

Frequency and repeated observations

Does an agreement fraction of $4/6$ establish a long-run agreement rate for future tickets?

No. A frequency account connects probability to relative frequency in a specified repeatable process. If I_i indicates agreement on trial i, the observed frequency after n trials is $n^{-1} \sum_{i=1}^{n} I_i$; a limiting account relates this sequence to a probability p. Six observations alone establish no such limit and do not establish that future tickets follow the same process.

Belief and evidence

Suppose an analyst assigns probability 0.8 to agreement on the next ticket, given this audit and other information. Does the table alone determine 0.8?

No. A subjective account treats probability as a degree of belief given information. The assignment needs stated information or judgments and must be coherent with the analyst's other probability assignments. An evidential account also needs a specified rule connecting evidence and background assumptions to a number.

What the earlier views contribute

What does each account emphasize: equal-likelihood counting, frequency, belief, or evidential support? What remains necessary for a calculation?

The accounts emphasize symmetry, repeated observations, information-conditioned belief, and evidential support. A calculation still needs specified outcomes, an event and a probability assignment. A propensity interpretation instead locates chance in a specified physical or randomized mechanism.

A second question: a random chord

A chord is selected at random from a circle. What is the probability that its length exceeds the side of an inscribed equilateral triangle?

The phrase “at random” does not say how the chord is generated. Bertrand’s chord problem has answers $1/3$, $1/2$ and $1/4$ under different sampling procedures. We will pose the question here and return to its full resolution in the supplement.

What must a probability model specify?

The ticket audit and the chord question both involve uncertainty. What mathematical objects would let us state the outcomes, the event and its probability?

Kolmogorov's set-based framework specifies a sample space Ω, an event collection $\mathcal{F}$, and a probability measure P. We now build these objects from the ticket-selection experiment.

Specify the ticket experiment

The six records are fixed. We select one ticket ID uniformly. What varies, and what are the possible outcomes?

The selected ID varies. The sample space is $\Omega = \{t_1, t_2, t_3, t_4, t_5, t_6\}$. The ticket contents and labels are fixed data in this experiment.

Build the Billing event

Which ticket IDs have Billing in the reviewer's reference outcome? Write them as a set B.

$B = \{t_1, t_2, t_5\}$. An event is a set of outcomes in the sample space. The statement "Billing is required" occurs exactly when the selected ID belongs to B.

Build the Account Support event

Which IDs have Account Support in the reviewer's reference outcome? Write this event as A.

$A = \{t_1, t_3, t_6\}$. The two events overlap because t_1 requires both destinations.

Other as a complement

In this two-destination question, “Other” means neither Billing nor Account Support is required. Express this event using A and B.

$O = (B \cup A)^c = \{t_4\}$. The complement is taken relative to Ω. It does not mean the ticket is unreviewed or unknown; another team may handle it.

Union and intersection

Use the ticket IDs to find $B \cap A$ and $B \cup A$. Interpret each event in words.

$B \cap A = \{t_1\}$ means both destinations are required. $B \cup A = \{t_1, t_2, t_3, t_5, t_6\}$ means at least one is required. “Or” in a union is inclusive.

Count the union without double-counting

Why is the number of IDs in $B \cup A$ not $|B|+|A|$? Calculate the probability under uniform selection.

The shared ID t_1 appears in both marginal counts. Split the union into disjoint parts: $B \backslash A = \{t_2, t_5\}$, $A \backslash B = \{t_3, t_6\}$, and $B \cap A = \{t_1\}$. Hence $P(B \cup A) = (2 + 2 + 1)/6 = 5/6$. Equivalently, $3/6 + 3/6 - 1/6 = 5/6$.

The probability-space triplet

For uniform selection from the six ticket IDs, specify the sample space, the events and the probability assignment.

$$(\Omega, \mathcal{F}, P)$$

$\Omega = \{t_1, t_2, t_3, t_4, t_5, t_6\}$.

$\mathcal{F} = 2^{\Omega}$, the collection of all subsets of Ω.

For every $E \in \mathcal{F}$, $P(E) = |E|/6$.

Thus $P(B) = 3/6 = 1/2$ and $P(B \cup A) = 5/6$.

Outcomes, events and probability values

Classify each object: t_2, B, and $P(B) = 1/2$. Which one is the outcome? Which is an event? Which is a number?

t_2 is an outcome in Ω. B is an event, a subset of Ω. $P(B) = 1/2$ is a number assigned to that event. The measure takes events as input.

Kolmogorov's probability axioms

The four audit categories have probabilities $2/6, 2/6, 1/6, 1/6$. They do not overlap and their probabilities sum to 1. What general rules does this suggest?

For $P: \mathcal{F} \to [0, 1]$:

1. **Nonnegativity:** $P(E) \geq 0$ for every $E \in \mathcal{F}$.
2. **Normalization:** $P(\Omega) = 1$.
3. **Countable additivity:** for pairwise disjoint $E_1, E_2, \ldots \in \mathcal{F}$, $P(\bigcup_{n=1}^{\infty} E_n) = \sum_{n=1}^{\infty} P(E_n)$.

For example, B and B^c are disjoint and exhaustive, and $P(B) + P(B^c) = 3/6 + 3/6 = 1$.

A random variable for Billing

Can we record numerically whether Billing is required? Define X_B on the ticket outcomes.

The values of X_B on $t_1, ..., t_6$ are $1, 1, 0, 0, 1, 0$. A random variable assigns a number to each outcome. Here $P(X_B = 1) = 3/6 = 1/2$. The function takes a ticket as input; the probability measure takes a set of tickets as input. Once the ticket is known, its recorded value is fixed.

Image of a set of outcomes

For $A = \{t_1, t_3, t_6\}$, what values does X_B produce? Write the image $X_B(A)$.

$X_B(A) = \{X_B(t) : t \in A\} = \{0, 1\}$. The image is a set of real values, not a set of ticket IDs.

Inverse image and probability

Which outcomes produce the value 1? Express them as $X_B^{-1}(\{1\})$ and calculate $P(X_B = 1)$.

$X_B^{-1}(\{1\}) = \{t \in \Omega : X_B(t) = 1\} = B = \{t_1, t_2, t_5\}$. Therefore $P(X_B = 1) = P(B) = 3/6 = 1/2$. The inverse image is an event; it need not be an inverse function.

The law of the Billing indicator

The Billing indicator is 1 on t_1, t_2, t_5 and 0 otherwise. Under uniform ID selection, what probability law does it induce?

The distribution gives the probabilities of the numerical values: $P(X_B = 1) = 1/2$ and $P(X_B = 0) = 1/2$. These probabilities sum to 1. Count the outcomes producing each value.

One die: outcome, event, value, probability

Roll a stipulated fair die. Let $\Omega_d = \{1, \ldots, 6\}$, $\mathcal{F}_d = 2^{\Omega_d}$, $P_d(E) = |E|/6$, and $Z(\omega) = \omega$. Distinguish 4, $\{4\}$, $Z(4)$ and $P_d(\{4\})$.

The outcome is 4; the singleton event is $\{4\}$; the function value is $Z(4) = 4$; its probability is $P_d(\{4\}) = 1/6$. Also $Z^{-1}(\{2, 4, 6\}) = \{2, 4, 6\}$, so $P_d(Z \text{ is even}) = 1/2$.

Worked check: neither named destination

Under uniform selection from the six audited IDs, find the probability of “Other,” meaning neither Billing nor Account Support is required.

$O = (B \cup A)^c = \{t_4\}$. Thus $P(O) = 1/6$.

Worked check: exactly one destination

Find the event and probability that exactly one of Billing and Account Support is required.

$$(B \backslash A) \cup (A \backslash B) = \{t_2, t_3, t_5, t_6\}.$$

Therefore its probability is $4/6 = 2/3$.

Worked check: exact agreement

Let C be the event that the model classification equals the reviewer reference outcome. Find C and its probability in the uniform ticket-ID experiment.

$C = \{t_2, t_3, t_5, t_6\}$, so $P(C) = 4/6 = 2/3$. This equals the observed agreement fraction because the experiment selects one ticket ID uniformly from this exact six-ticket audit.

Worked check: agreement as a random variable

Define $Y(t) = 1$ for exact agreement and 0 otherwise. Find $Y^{-1}(\{1\})$ and $P(Y = 1)$.

$Y^{-1}(\{1\}) = C = \{t_2, t_3, t_5, t_6\}$. Thus $P(Y = 1) = P(C) = 4/6 = 2/3$. The random variable records an outcome numerically; its inverse image is the event; the measure assigns that event a number.

Weighted audit selection

Keep the six audit records, but select IDs with weights $(1, 1, 1, 3, 1, 1)/8$. Agreement is $C = \{t_2, t_3, t_5, t_6\}$; Other is $\{t_4\}$. Find both probabilities.

The weights sum to one and define $Q(E) = \sum_{t_i \in E} w_i$. Therefore $Q(C) = 4/8 = 1/2$ and $Q(O) = 3/8$. The observed fraction remains $4/6$; the changed experiment gives a different selection probability.

Fraud-screen worked example

In a hypothetical batch of 20 transactions, reviewers mark 4 as fraud. A screen flags 3 of those and 2 of the other 16. Select one ID uniformly. Find the false-positive and false-negative probabilities.

With F the flagged set and H the reviewer-positive set, $P(F \cap H^c) = 2/20 = 0.10$ and $P(F^c \cap H) = 1/20 = 0.05$. Selecting instead from the 16 reviewer-negative IDs gives false-positive fraction $2/16 = 0.125$. State the population before choosing a denominator.

Sensor-tolerance worked example

A hypothetical sensor error ε is uniform on $[-2, 2]$ mm. Its density is $1/4$ on that interval. Find the probability that the error exceeds 1 mm in absolute value.

The event is $[-2, -1) \cup (1, 2]$, of total length 2. Thus $P(|\varepsilon| > 1) = 2/4 = 1/2$. The density is $1/4$ per mm on $[-2, 2]$. Probability is area; the total area is 1. Uniform error is stipulated.

Assessment-indicator worked example

Select one of five hypothetical records uniformly. Scores are 45, 60, 70, 80 and 95; passing means at least 60. Define the pass indicator and its law.

On record IDs $a_1, \ldots, a_5$, let $Y(a_i) = 1$ for a passing score and 0 otherwise. Then $Y^{-1}(\{1\}) = \{a_2, a_3, a_4, a_5\}$, so $P_Y(\{1\}) = 4/5$ and $P_Y(\{0\}) = 1/5$. This describes selection from these records, not a future cohort.

Electrical engineering: two power supplies

A hypothetical device has a primary and a backup supply. The stipulated probabilities for (primary, backup) are (W, W): 0.90, (W, F): 0.06, (F, W): 0.03, (F, F): 0.01, where W means working and F means failed. The device operates if at least one supply works. Find its operating probability.

The four states are the outcomes; all their subsets are events. Their probabilities are nonnegative and sum to 1. The device fails only in (F, F), so $P(\text{operates}) = 1 - 0.01 = 0.99$. Direct addition gives $0.90 + 0.06 + 0.03 = 0.99$. This introduces the complement rule $P(E^c) = 1 - P(E)$. The joint state probabilities are given; independence of the supplies is not assumed.

IT operations: overlapping incident tags

Select uniformly one of 40 fixed incident IDs in a hypothetical log. Twelve have a network tag, nine a storage tag, and four have both. Find the probabilities of at least one tag, exactly one tag, and neither tag.

Let N and S be the two tagged sets. At least one contains $12 + 9 - 4 = 17$ IDs, so its probability is $17/40$. Exactly one contains $(12 - 4) + (9 - 4) = 13$ IDs, giving $13/40$. Neither contains $40 - 17 = 23$ IDs, giving $23/40$. Tags can overlap even though only one incident ID is selected.

Accounting: reconciliation categories

A hypothetical ledger has 30 distinct invoice IDs: 21 reconcile correctly, 5 have only a missing receipt, 3 have only an amount mismatch, and 1 has both issues. Select one ID uniformly. Find the probabilities of any issue and of an amount mismatch.

The four disjoint categories have probabilities $21/30, 5/30, 3/30, 1/30$, which sum to 1. Any issue has probability $(5 + 3 + 1)/30 = 9/30$. An amount mismatch includes the last two categories, so its probability is $(3 + 1)/30 = 4/30$. Category names must specify whether overlapping issues are included.

Agriculture: plot selection and area selection

A hypothetical farm has plot A of area 1 hectare and plot B of area 3 hectares. All of A is classified as stressed and none of B is. First select a plot ID uniformly and report its classification. Then consider selecting a location uniformly by area across the farm. Find the probability of reporting stress in each experiment.

For uniform plot selection, $\Omega = \{A, B\}$ and $P(\{A\}) = 1/2$. For uniform area selection, the stressed area is 1 out of 4 hectares, giving probability $1/4$. At the plot-label level the second experiment assigns weights $1/4$ and $3/4$. Changing the sampling unit changes the probability. Neither answer estimates an unstated future drought risk.

Finance: a loss distribution from fixed records

Select one of ten hypothetical contract records uniformly. Six have a recorded loss of ₹0, three have ₹100, and one has ₹500. Let L record the loss. Find its distribution, $P(L > 0)$ and $P(L \geq 100)$.

The outcomes are the ten contract IDs, while L takes values 0, 100 and 500. Counting the IDs gives $P(L = 0) = 6/10$, $P(L = 100) = 3/10$, and $P(L = 500) = 1/10$. Thus $P(L > 0) = 4/10$ and $P(L \geq 100) = 4/10$. The events coincide on these records. Three loss values do not receive equal probabilities. This describes selection from a fixed ledger, not a forecast.

Support: number of required destinations

For the six-ticket audit, record $D(t)$, the number of named destinations required by the reviewer: Billing and Account Support. Find the distribution of D under uniform ticket selection.

Ticket t_4 gives 0, tickets t_2, t_3, t_5, t_6 give 1, and t_1 gives 2. Hence $P(D=0)=1/6$, $P(D=1)=4/6$ and $P(D=2)=1/6$. The probabilities sum to 1. “Other” means zero of these two named destinations, not that no support is needed.

Agriculture: germination counts

Select one of eight hypothetical seed-tray records uniformly. The recorded numbers of germinated seeds are $0, 1, 1, 2, 2, 2, 3, 3$. Let G be the recorded count. Find its distribution and the probability of at least two germinated seeds.

Tray IDs are equally likely; their counts are not. The distribution is $P(G=0)=1/8$, $P(G=1)=2/8$, $P(G=2)=3/8$, $P(G=3)=2/8$. Therefore $P(G \geq 2) = (3+2)/8 = 5/8$. No assumption of independent seed germination is needed to describe selection from these fixed records.

IT monitoring: a continuous waiting time

A monitoring check runs every 10 minutes. Stipulate that an incident arrives at a uniformly distributed time within one checking cycle. Let W be the wait until the next check. What is $P(W > 7)$?

The waiting time is uniform on $[0, 10]$ minutes. Waiting more than 7 minutes occupies an interval of length 3, so $P(W > 7) = 3/10$. Equivalently, the density is $1/10$ per minute and the area is $3(1/10)$. A periodic checking schedule alone does not establish the uniform arrival assumption.